pfi

# Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

# The Framework

WEEK 2 · DAY 4

Created by **Jeff Lenamon**

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# Today: the framework

### Days 2 to 8: your own code

| | |
|---|---|
| `tool_schema` | the docstring, read |
| `agent_loop` | request, tools, repeat |
| `run_tools` | the message bookkeeping |
| `check_figures` | every figure, checked |

### Today

1. What a framework is for
2. Tracing off, and why
3. function_tool beside tool_schema
4. sdk_agent on the course's client
5. Day 2's question both ways: body_diff
6. await in a notebook
7. sdk_trace_table beside trace_table
8. What it does not do
9. LangGraph in one paragraph

**Every model response in the notebook is a saved response, replayed. A live run can differ.**

# The rules for agents

## What an agent produces: text, every number from a tool, whoever builds the request

### Allowed names

the agent returned

the saved run

the brief states

### Never called

the agent's opinion or view

what the agent believes

the agent's advice

## On the synthetic book and the curve: numbers from your tools, never a view

### Allowed names

DV01 per 100 of face

position DV01 in dollars

the largest DTS contributions in the file

### Never called

cheap, rich, or attractive

a trade idea or a recommendation

overweight or underweight

# What a framework is for

| | Days 2 to 8 | The framework |
|---|---|---|
| Tool schemas | tool_schema | function_tool |
| The loop | agent_loop | Runner.run |
| The stop rule | max_steps | max_turns |
| Bookkeeping | run_tools, tool_message | its own items |
| Checks on the answer | check_figures, yours | none |

## Story: the package's own record

openai-agents on PyPI: 0.0.1 uploaded 2025-03-04, yanked ("Updating to real release"); 0.0.2 on 2025-03-11.

The course pins 0.23.1, uploaded 2026-10-02; PyPI lists 123 versions with files.

Sources (read 2026-10-11): pypi.org/pypi/openai-agents/json; api.github.com/repos/openai/openai-agents-python

**A framework writes four of the five parts. The fifth, the checks, stays yours.**

# Tracing off, and why

```
os.environ["OPENAI_AGENTS_DISABLE_TRACING"] = "1"   # before import agents: read once, at start
import agents
agents.set_tracing_disabled(True)                     # after: a manual setting wins from then on
```

## NoOpTrace

what a new trace is now

## two

switches, so import order cannot undo it

**The default exporter, read in agents/tracing/processors.py:**

`https://api.openai.com/v1/traces/ingest`

**Nothing about a run leaves the machine unasked, and the saved runs hold only the agent's requests.**

# function_tool beside tool_schema

### function_tool(dv01): the framework's schema


```
{"properties": {"settlement": {"description": "the settlement date as YYYY-MM-DD, for example 2026-10-02
.", "title": "Settlement", "type": "string"}, "maturity": {"description": "the maturity date as YYYY-MM-
DD.", "title": "Maturity", "type": "string"}, "coupon_pct": {"description": "the annual coupon in percen
t, 5.0 for a 5% coupon.", "title": "Coupon Pct", "type": "number"}, "yield_pct": {"description": "the yi
eld to maturity in percent, 5.06 for 5.06%.", "title": "Yield Pct", "type": "number"}}, "required": ["se
ttlement", "maturity", "coupon_pct", "yield_pct"], "title": "dv01_args", "type": "object", "additionalPr
```


### tool_schema(dv01): day 1's


```
{"type": "object", "properties": {"settlement": {"type": "string", "description": "the settlement date a
s YYYY-MM-DD, for example 2026-10-02."}, "maturity": {"type": "string", "description": "the maturity dat
e as YYYY-MM-DD."}, "coupon_pct": {"type": "number", "description": "the annual coupon in percent, 5.0 f
or a 5% coupon."}, "yield_pct": {"type": "number", "description": "the yield to maturity in percent, 5.0
6 for 5.06%."}}, "required": ["settlement", "maturity", "coupon_pct", "yield_pct"], "additionalPropertie
```


**The framework adds titles and nothing else. Equal once the titles are taken out: True.**

# sdk_agent on the course's client

```
desk_agent = agentkit.sdk_agent("desk", "Answer with the tool.", [dv01], aclient, agentkit.MODELS["desk"], 9)
```

```
instructions start with DESK_PROMPT: True; its own sentence: 'Answer with the tool.'
temperature 0.0, parallel_tool_calls False, store False, extra_args {'seed': 42, 'prompt_cache_key': 'cours...
tools ['dv01'], parameters sent ['settlement', 'maturity', 'coupon_pct', 'yield_pct']
```

**How the framework chooses store, in its own source:**

```
default_store = True if cls.is_openai(client) else None
return model_settings.store if model_settings.store is not None else default_store
```

**On OpenAI's own address, store is true unless the settings say otherwise; elsewhere it is not sent.**

# Day 2's question, both ways: body_diff

**body_diff(loop, framework)**

| field | in_a | in_b |
|---|---|---|
| messages | True | True |
| prompt_cache_key | False | True |
| store | False | True |
| tools | True | True |

```
store: False; prompt_cache_key: 'course8-day09'
system message: DESK_PROMPT + '\nAnswer with the tool.'
the user message is the same: True
the tools are the same once the titles are taken out: True
temperature as sent: the loop 0, the framework 0.0
the two first requests have the same exchange key: no
```

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 951 prompt and 104 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 1,008 prompt and 126 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

**Added: prompt_cache_key and store. Temperature 0 against 0.0: equal as numbers, other bytes, another key.**

# Why sdk_settings sets two fields

```
default_agent = agents.Agent(..., model=agents.OpenAIChatCompletionsModel(model=..., openai_client=fake_defaults))
await agents.Runner.run(default_agent, QUESTION, max_turns=8)        # twice, each in its own cell
```

```
run 1: store True; prompt_cache_key begins agents-sdk:run: True
run 2: store True; prompt_cache_key begins agents-sdk:run: True
the two runs sent the same prompt_cache_key: no
the same question, the same exchange key: no
requests to a traces path: 0
```

## store=False

the response is not kept

## course8-day09

the same key on every run

**A new random key per run changes every body, so every exchange key: a saved run is never found again.**

# await Runner.run in a notebook

```
outcome = await agents.Runner.run(desk_agent, QUESTION, max_turns=8)   # a notebook: one await per cell
asyncio.run(main())                                                    # a script of your own
Runner.run_sync(...)        # the scan refuses it in a notebook; its source says it cannot run here
```

```
asyncio.run: asyncio.run() cannot be called from a running event loop
requests sent by this cell: 0
```

## coroutine

a run described, not started

## one await

per cell: one failed request, one cell

**A notebook already runs an event loop: await hands the run to it.**

# sdk_trace_table beside trace_table

```
the loop, trace_table:
 step role tool arguments finish_reason prompt_tokens completion_tokens
 0 user <NA> <NA>
 1 assistant dv01 {...} tool_calls 432 42
 1 tool dv01 <NA> <NA>
 2 assistant stop 519 62
the framework, sdk_trace_table:
 step role tool arguments finish_reason prompt_tokens completion_tokens
 0 user <NA> <NA>
 1 assistant dv01 {...} tool_calls 459 42
 1 tool dv01 <NA> <NA>
 2 assistant stop 549 84
the same columns: True
```

**The same columns, read from the bodies the client sent: every check of the course works on both.**

# What the framework does not do: check a number

```
market_agent = agentkit.sdk_agent("market desk", ..., output_type=agentkit.DeskAnswer)
agentkit.check_figures(outcome_market.final_output, run_market)
```

```
Change in 10 Yr par yield between 2026-09-16 and 2026-10-02 27.0 bp 27.0 True True True True
2 Yr to 10 Yr slope on 2026-10-02 45.0 bp 45.0 True True True True
```

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 3 requests, 3,643 prompt and 262 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

```
loop       numbers no tool returned: []
framework  numbers no tool returned: []
```

**The framework checks the JSON's shape. Whether a figure is its tool's value: your code, after the run.**

# LangGraph, Excel, and Git

## LangGraph, as the course probed it on 2026-10-08

langgraph 1.2.14, langchain-core 1.6.7: MIT, released 2026-10-06, about 19.7 MB

Its nodes are your code: request bodies byte-identical to the raw loop's; replayed as shipped

LangSmith tracing off unless its variables are set. Not installed in the course.

**Excel: no counterpart today. No agent framework, no tool requests; today's numbers were tested on days 1, 5, 6, 8.**

```
agentkit.py | 142 ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
test_agentkit.py | 70 ++++++++++++++++++++++++++
2 files changed, 212 insertions(+)
```

**git diff --stat HEAD~1 HEAD in the notebook; git diff --stat week1 HEAD in my_bondmath. 47 passed.**

# The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

## Docstring

Tracing off, store false, the fixed key course8-day09.

2

## Code back

Written for the course in the style of an AI answer. It answers.

3

## Test first

No trace processor, a fake server, one run. It fails on store.

4

## Fix

Tracing off; sdk_agent with sdk_settings inside. It passes.

**Today: make_agent switches tracing on and leaves the defaults: store true, a random cache key every run.**

*Code that runs is not the same as code that is right.*

# What you can do now

- A framework writes schemas, the loop, the stop rule; not the checks
- Tracing off, twice, before and after import agents
- body_diff: the framework adds store and prompt_cache_key
- sdk_settings: store false, a fixed key, so runs replay
- One await per cell; check_figures on the framework's run

**Week 2, Day 5** Case study: a macro agent on two statements